

Abstract

Different batches of single-cell RNA sequencing data are often collected at different times, from different labs, and/or using slightly different procedures. This can lead to systematic differences between various batches of data called the batch effect.

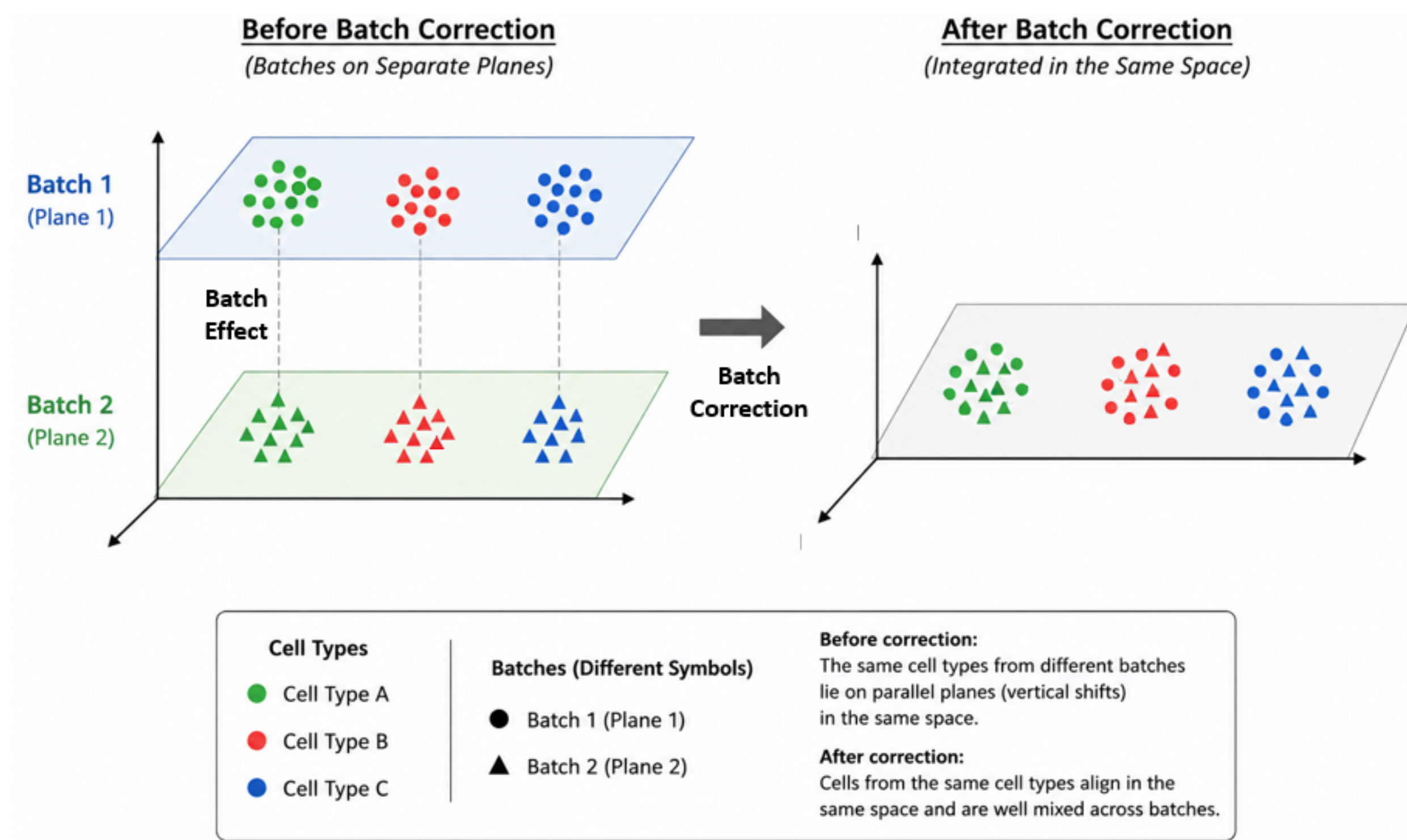


Figure 1. Illustration of batch effect and correction.

This project investigates different methods for detection and correction of the batch effect, while preserving biological information, so that batches may be integrated into one large dataset. If the batch effect is not corrected for, there may be unwanted variation in the integrated dataset that is not due to biological signal. This additional noise could cause issues in downstream analysis as it might be mistaken for real signal.

Batch Correction Methods

The three batch correction methods that were compared in this project are described as follows:

- Matching mutual nearest neighbours (MNN) [1].** Cells across batches who are among each other's k nearest neighbours are paired. These points are referred to as anchors and are believed to be biologically similar. The difference between anchor pairs should mainly be caused by the batch effect, and these difference vectors can be used to approximate the batch effect. These approximations are then used to construct a correction map.
- Fast mutual nearest neighbours (FastMNN) [2].** Builds on the MNN method, but PCA is performed ahead of time and matching is performed in the PCA embedding space.
- Seurat's integration method with canonical correlation analysis (Seurat CCA) [3].** Similar to fastMNN except CCA is performed on the two batches to get the low-dimensional embedding rather than PCA. The advantage is that CCA seeks an embedding that preserves shared structure between the datasets.

Geometry of the Batch Effect

Preliminary analysis suggested that the batch effect consists of biological distortion and technical shift components. The metrics developed in what follows attempt to quantify the magnitude of these two components before and after integration has been applied.

Each cell is treated as a point x in $\mathbb{R}^k$, where k is the dimension of the embedding space. Define the **relative distance** between cell-types t and s in batch b as

$$\mu_{s,t}^{(b)} = \frac{\text{mean} \left\{ \|x - y\|_2 : x \in C_s^{(b)}, y \in C_t^{(b)} \right\}}{\text{median} \left\{ \|x - y\|_2 : x \neq y, x, y \in C_s^{(b)} \right\}}.$$

where $C_t^{(b)}$ is the set of all cells of type t in batch b . In what follows, the reference cell-type is chosen to be $s = \alpha$. In this way, the distances are always computed relative to the "size" of the α cluster.

The metric $\Delta_{\alpha,t}^{(b_1,b_2)} = \mu_{\alpha,t}^{(b_1)} - \mu_{\alpha,t}^{(b_2)}$ is used to compare this geometry across batches. Positive $\Delta_{\alpha,t}^{(b_1,b_2)}$ means t is farther from α in b_1 ; negative means farther in b_2 ; values near zero indicate similar α -to- t geometry, this would be ideal after correction.

Define the **technical mixing ratio** for cell-type t across batches b_1 and b_2 as

$$R_t^{(b_1,b_2)} = \frac{\text{median} \left\{ \|x - y\|_2 : x \in C_t^{(b_1)}, y \in C_t^{(b_2)} \right\}}{\text{median} \left(\bigcup_{i=1,2} \left\{ \|x - y\|_2 : x, y \in C_t^{(b_i)} \right\} \right)}.$$

For each correction method m , the median $R_t^{(b_1,b_2)}$ is investigated for all combinations of t , b_1 , and b_2 . Under ideal circumstances, this value should be close to one as this indicates good mixing.



Figure 2. Heatmap of technical mixing ratios across cell-types and batch pairs.

The **biological preservation** metric for each correction method m is

$$B^{(m)} = \text{median}_{b,t} \left| \log \left(\mu_{\alpha,t}^{(m,b)} / \mu_{\alpha,t}^{(\text{PCA},b)} \right) \right|.$$

Lower values mean that the corrected embedding changed the original α -centered biological geometry less relative to the uncorrected data. This being close to zero indicates preservation of the original biological structure.

The **batch agreement** metric $A^{(m)}$ for method m is the median of $|\Delta_{\alpha,t}^{(b_1,b_2)}|$ measured after integration using method m across all values of t , b_1 , and b_2 . Lower values mean batches agree more closely on α -centered geometry, but this is only good if biological separation is still preserved.

Method	median R	$A^{(m)}$	$B^{(m)}$
Seurat CCA	1.0069	0.9727	0.2258
FastMNN	1.0530	0.2156	0.6704
MNN	1.0500	0.3351	0.6218
Uncorrected PCA	1.7981	0.9196	0.0000

Table 1. Metrics used to compare batch correction methods.

Comparison of Integration Methods

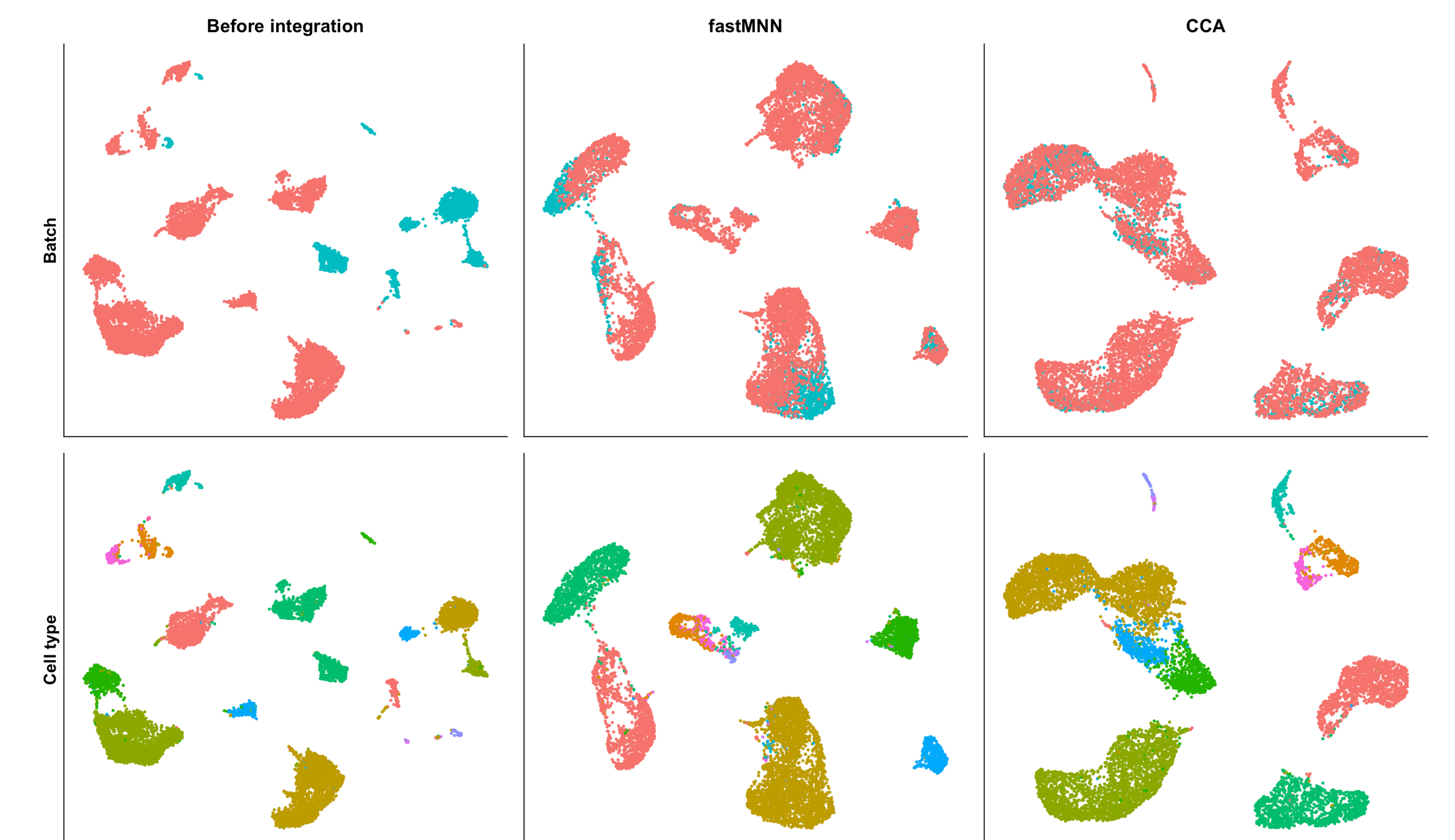


Figure 3. UMAP embeddings before and after correction. The top row of UMAP plots are coloured by batch, where coral indicates indrop and teal indicates smartseq2. The bottom row of UMAP plots is coloured by cell-type to show whether biological structure is preserved. The first column is before integration, the second column is fastMNN, and the third column is Seurat CCA.

Ideally, the clusters are expected to be well mixed when coloured by batch, and very distinct when coloured by cell-type. This would indicate that both the biological signal has been preserved and the batch effect has been corrected.

Conclusions

It was observed that Seurat CCA had the best median technical mixing ratio, followed by FastMNN and MNN. This is reflected in the UMAPs as well, with Seurat CCA visually having the most well-mixed clusters in terms of batch type. However, Seurat CCA performed worse in terms of batch agreement compared to FastMNN. This agrees with the UMAPs, as the FastMNN clusters are a bit more distinct in terms of cell-type than the Seurat CCA clusters.

References

- Laleh Haghverdi, Aaron T. L. Lun, Michael D. Morgan, and John C. Marioni. Batch effects in single-cell rna-sequencing data are corrected by matching mutual nearest neighbors. *Nature Biotechnology*, 36(5):421–427, 2018.
- Aaron T. L. Lun. A description of the theory behind the fastmnn algorithm. <https://marionilab.github.io/FurtherMNN2018/theory/description.html>, 2019. Accessed: 2026-06-17.
- Tim Stuart, Andrew Butler, Paul Hoffman, Christoph Hafemeister, Efthymia Papalexi, William M. III Mauck, Yuhao Hao, Marlon Stoeckius, Peter Smibert, and Rahul Satija. Comprehensive integration of single-cell data. *Cell*, 177(7):1888–1902.e21, 2019.